

Examiner reports

Q1.

Students almost always scored full marks for their answers to parts (a)(i) and (a)(ii) but writing the n th term of the geometric series in the required form in (a)(iii) proved to be much more challenging. Although the majority of students started correctly by writing $420 \times 0.7^{n-1}$, a large proportion of these students failed to write this in the form 600×0.7^n . The most common wrong answer, for those who attempted to re-write $420 \times 0.7^{n-1}$ in the stated form, was 294×0.7^n . In part (b)(ii) there was some confusion with the term and sigma notation which resulted in the sum being equated to zero. The most successful approach was to solve $U_k = 0$ to obtain the value of the positive integer k and then to find

the value of $\frac{k}{2} (240 + 0)$, which represents $\sum_{n=1}^k u_n$, the sum of the k terms of the arithmetic series with 1st term 240 and last term $0 (= U_k)$. In a minority of cases where the correct expression for U_k was equated to zero, an incorrect rearrangement resulted in a negative value for k . Students who made this error did not seem to realise that such a value for k was meaningless.

Q2.

Part (a) was a good source of marks for most candidates, with the formula for the sum of an arithmetic series being well understood and the necessary stage(s) needed in proving a printed answer being appreciated. Part (b) was also well answered, though a number of candidates wrongly used u_5 as $a + 5d$ or, worse still, used the formula for S_5 , the sum to five terms. Not all candidates who obtained the two correct linear equations were able to correctly solve them simultaneously.

Part (c), as expected, proved to be the least well answered part of the question, with a number of candidates making no attempt to answer it. Those candidates who failed to

replace $\sum_{n=1}^{25} u_n$ by 3500 rarely made any progress. The most common error was to

replace $\sum_{n=1}^{25} u_n$ by 325, presumably obtained by applying the irrelevant formula

$\sum_{r=1}^n r = \frac{1}{2}n(n+1)$ which appears in the formulae booklet.

Q3.

It was pleasing to see a very high proportion of candidates showing sufficient detail in their solutions to completely justify their obtaining of the printed result in part (a). Most candidates used the correct general formula for the n th term of an arithmetic series to find u_{16} and u_{21} in part (b), but it was not uncommon to see the multiplier 2 being applied to u_{21} instead of u_{16} . The final part of the question was answered poorly with many candidates

either solving $u_k = 0$ instead of $S_k = 0$, or looking in the formulae booklet and writing $\frac{k}{2} (k + 1) = 0$.

Q4.

Many candidates correctly showed the printed value for r in part (a)(i) by forming and

solving the equation ' $\frac{10}{1-r} = 50$ '. Those who used verification methods rarely scored both marks. Most candidates correctly found the second term of the geometric series, but a significant number of candidates in part (b)(i) tried to 'guess' the answer for the common difference instead of using the formula for the n^{th} term of an arithmetic series and ended up with the wrong answer ' $d = 0.5$ '.

The final part of the question, which involved understanding of the sigma sign, was better answered than in previous papers but still the majority of candidates failed to obtain the

correct value. The most common wrong answer was 820, obtained by using $\frac{1}{2}n(n+1)$ with $n = 40$, presumably from scanning through the formulae booklet.

Q5.

Part (a) was normally answered correctly but those who relied entirely on a verification approach generally scored no more than 1 mark and many of these scored no marks because they failed to state a conclusion.

Most candidates were able to find the correct values for the first term and sum to infinity of the geometric series but many could make no progress in the final part of the question which involved the sigma sign. Of the better solutions, the very common error was to find the value of ' $S_{\infty} - S_6$ '. The most common correct methods involved either finding the value of ' $S_{\infty} - S_5$ ' or recognising that the sum to infinity of a geometric series with first term $u_6 (= 48.6)$ and common ratio 0.6 was required.

Q6.

The majority of candidates presented correct solutions to parts (a) and (b) although the value 36 for the common ratio appeared more often than expected. In part (c)(i), although many candidates applied the correct formula for the sum to 12 terms of the geometric sequence, others focused on the form of the printed answer and replaced n in the general formula by k . Most of these candidates failed to gain any marks for this part because they did not use $n = 12$. A very common error was ' $-4(1 - 4^{12}) = -4 + 16^{12}$ ' which resulted in the answer ' $k = 24$ '. Sign errors were also common. In general, only the most able candidates were able to see how to use the earlier parts to answer part (c)(ii) and even some of these failed to evaluate their expression to reach the correct value 67108848.

Q7.

Part (a) was generally well answered although some tried to use formulae for geometric series rather than arithmetic series. Most were able to find the sum to 20 terms in part (b) although arithmetical errors were noted. Part (c), which involved sigma notation, was not always attempted. The common errors seen otherwise were subtracting S_{21} rather than S_{20} and, normally only produced by the weaker candidates, using the formula for the n^{th} term instead of the sum to n terms. Some candidates wasted time by showing that the sum to 50 terms was indeed the given value, 11525.